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Team 04 Final Presentation

Robot Programing with ROS  
Summer Semester 2026

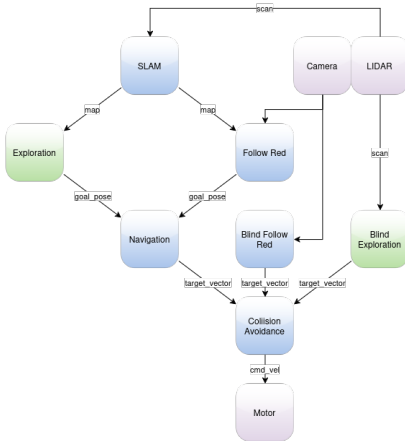
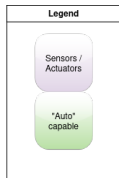
Jun 03, 2026

# Outline

- 1 Overview
- 2 Exploration
- 3 Collision Avoidance
- 4 Navigation
- 5 Follow Red
- 6 Manager
- 7 Navigation (stuck, smoothing,  $\theta^*$  v  $a^*$ )
- 8 Localization (in Sim)

# Overview

# Overview



# Exploration

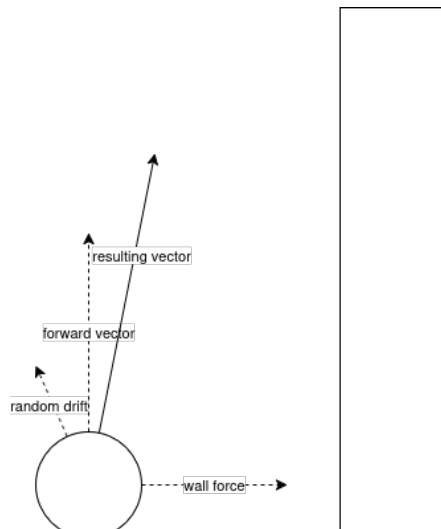
# Blind Exploration

- No map given

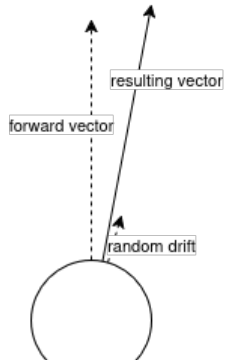
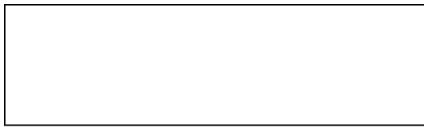
To complete the main goal "*Just keep moving!*" we do this:

- ➊ Apply random drift
- ➋ Drive forward
- ➌ Wall detected?
  - ➊ Yes: apply force to the side of the wall
- ➍ Haven't moved for a while?
  - ➊ Yes: apply randomized rotation

# Blind Exploration



## Blind Exploration - Why the drift?



A direction is picked through the random drift

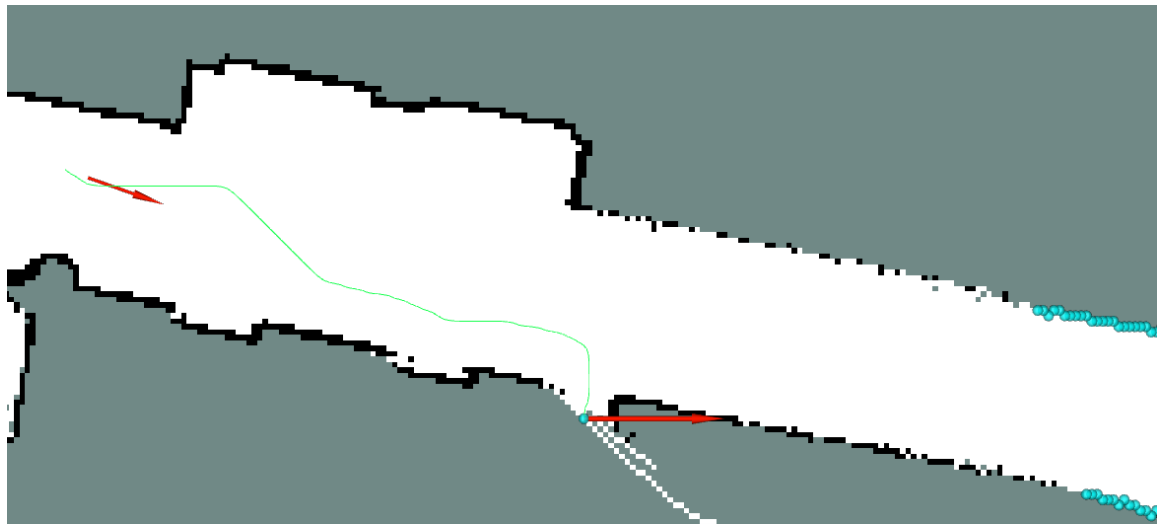
# Exploration

- Sufficient exploration has taken place
- Map given

We now know where we are and where we've been, the goal changes to "*Explore the unknown!*":

- ① Flood fill the map from our current position
- ② Create a frontier map
- ③ Select the closes frontier position (we haven't visited yet)
- ④ Send desired location to `/goal_pose`
- ⑤ Mark position as visited

# Exploration

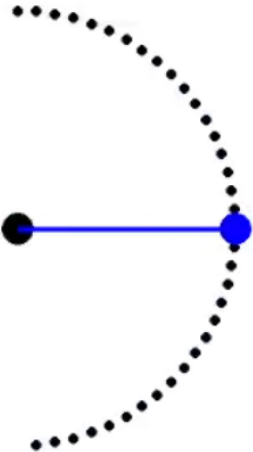


# Collision Avoidance

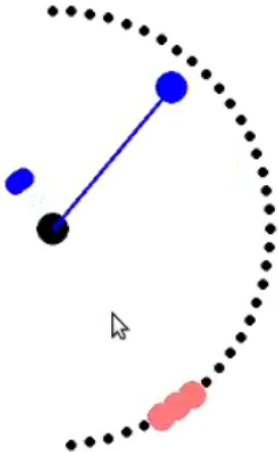
# Collision Avoidance

- Has to be active if the robot wants to move
- Receives an `/target_vector` with a linear and angular velocity
- Ensures...
  - that the velocities are within safe bounds
  - that the movement direction does not point into obstacles
- Converts `/target_vector` to `/cmd_vel` and sends it off to motor

# Collision Avoidance



# Collision Avoidance



# Navigation

# Navigation Steps:

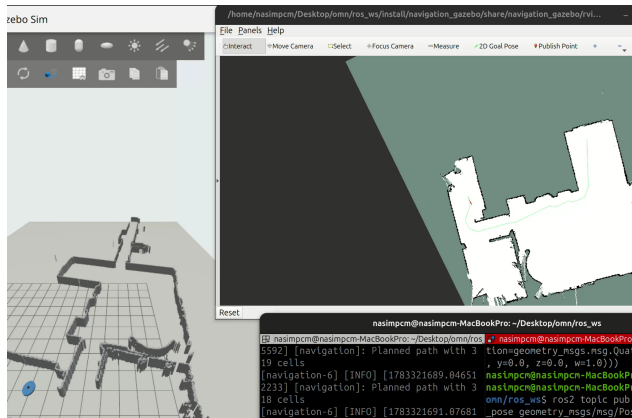
## Planning

- ➊ `timer_callback()` waits until a map, pose, and goal are available.
- ➋ Robot and goal positions are converted from metres to map-grid cells.
- ➌ A\* plans a path using horizontal, vertical, and diagonal moves.
- ➍ Obstacles are inflated to maintain clearance, and the grid path is smoothed to reduce zig-zag motion.

## Path Following

- ➎ Every 0.2 s, a waypoint approximately 0.4 m ahead is selected.
- ➏ A `TargetVector` pointing toward the waypoint is published.
- ➐ When the goal is within 0.4 m, the robot is stopped and `goal_succeeded` is reported.

# A\* Path Planning



## Evaluation Function

$$f(n) = g(n) + h(n)$$

$g(n)$  is the path cost from the start and  $h(n)$  estimates the remaining cost to the goal.

- Eight-connected grid search
- Straight-move cost: 1
- Diagonal-move cost:  $\sqrt{2}$
- Corner-cutting is prevented
- Inflated cells add obstacle-clearance cost

# A\* Implementation

- ➊ **Initialize:** add the start cell to the open-set priority queue, set  $g(\text{start}) = 0$ , and initialize `came_from`, `g_score`, and the closed set.
- ➋ **Select:** remove the cell with the smallest estimated total cost  $f(n)$ .
- ➌ **Check the goal:** if selected, reconstruct the path from the stored parent cells.
- ➍ **Mark explored:** skip cells already processed; otherwise add the cell to the closed set.
- ➎ **Generate neighbors:** consider all eight adjacent cells using straight and diagonal move costs.
- ➏ **Apply obstacle costs:** reject blocked or invalid cells and penalize inflated cells near obstacles.
- ➐ **Update better paths:** if a lower cost is found, update the neighbor's parent and scores, then add it to the open set.
- ➑ **Finish:** return the reconstructed path, or report `no_path` if the open set becomes empty.

## Next Waypoint Selection

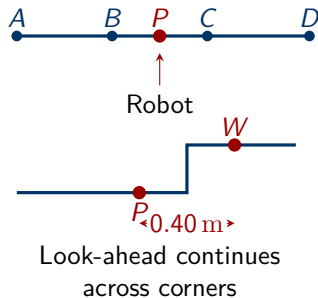
### 1. Locate the Robot on the Path

Project the robot onto every remaining path segment and select the closest point  $P$ .

### 2. Look Ahead Along the Path

Starting at  $P$ , measure 0.40 m along the path to obtain waypoint  $W$ . If the distance crosses a corner, continue on the next segment.

- `path_progress` prevents the selected position from jumping backward when the path passes close to itself.
- If less than 0.40 m remains, return the final goal point



## Follow Red

## Goal in 3 Steps

- 1 Detect a red cube via OpenCV
  - 2 Estimate its distance/angle or global map position
  - 3 Publish targets in a ROS 2 system
- **Node Type:** ROS 2 Lifecycle Node
  - **Input:** OpenCV camera stream
  - **Processing:** HSV masking, exponential smoothing, median filtering
  - **Output:** Polar coordinates (TargetVector) or global map poses (PoseStamped)

## Given Hardware and Calibration

|                     |         |
|---------------------|---------|
| Type                | Webcam  |
| Frame               | 640x480 |
| FOV                 | 60      |
| Calibrated Distance | 1 Meter |
| Calibrated Pixels   | 9750    |

## Dual Target Publishing Modes

Controlled by the runtime parameter `send_polar`:

### Polar Mode (`send_polar = True`)

- Publishes `TargetVector` to `/target_vector`.
- Direct values for linear (distance) and angular (bearing angle).

### Global Pose Mode (`send_polar = False`)

- Uses `tf2_ros` for transform lookups (`map`  $\rightarrow$  `base_footprint`).
- Transforms camera relative coordinates into global map coordinates.
- Publishes `PoseStamped` to `/goal_pose`.

# Image Processing & Filter Pipeline

## 1 Filter Size:

- Accept only a sum over 500px as possible cube.

## 2 Area Change Rate Filter (Jump Filter):

- Ignores frames with sudden, extreme area size spikes ( $\frac{A_{new}}{A_{old}} \notin [\min, \max]$ ).
- Prevents position jumps caused by brief reflections or partial occlusions.

## 3 HSV Masking:

- Dual range masking for red hue due to HSV wrap-around ( $0^\circ - 5^\circ$  and  $174^\circ - 179^\circ$ ).

## 4 Low-Pass Filter (Exponential Smoothing):

- Noise reduction for pixel position ( $\alpha_{pos} = 0.25$ ) and area ( $\alpha_{area} = 0.20$ ).

## 5 Median Ring Buffer:

- `deque(maxlen=5)` eliminates sporadic false detections over 5 frames.

## 6 Distance Estimation:

- Inverse square-root calculation based on calibrated pixel area:

$$d = d_0 \cdot \sqrt{\frac{A_0}{A}}$$

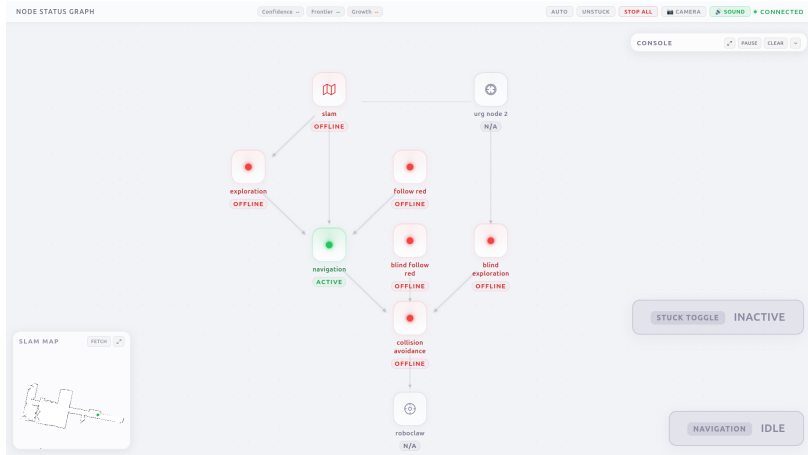
# Fault Tolerance

## Automatic Camera Reconnect

- Monitors consecutive read errors
- When reaching CAMERA\_MAX\_ERRORS (40 frames  $\approx$  2 s), the video device is released and re-initialized.

# Manager

# Dashboard



## Navigation (stuck, smoothing, $\theta^*$ v $a^*$ )

# Path Smoothing via LOS

## LOS-Algorithm

- Path from A\* gets checked for shortcuts
- Line of Sight is used
- starts from first point, checks for 50 cells ahead
- looks at furthest away, moves closer until visible
- repeat for the next point

## Reasons for/against using

- **For:** Very smooth path
- **For:** Decreased number of waypoints
- **For:** Changable Look-Ahead-Distance
- **Against:** High computational Cost
- **Against:** Too slow for constant path replanning
- Result: Not used

# Path Smoothing via Gradient

## Algorithm

- Path from A\* gets corrected
- Multiple passes
- each coordinate gets tied to its origin
- and gets pulled away by:  $y_{next} + y_{prev} - 2.0 * y_{curr}$
- stops automatically after specified passes or little change occurring

## Reasons for/against using

- **For:** Low Computational Cost
- **For:** Tweakable
- **Against:** Path not as smooth
- **Against:** Might drag path through walls
- **Result:** Used

# Stuck Logic

## Algorithm

- only when in mode "tracking" to avoid timeout while calculating path
- checks whether the robot has moved enough in a specified timeframe
- when stuck, clears the current path and publishes velocities = 0
- when next\_goal in queue, switch over

## Toggle Logic

set toggle to false, when:

- goal is reached or new current goal detected
- stuck and there is a next goal
- path is cleared, there is no path or a new path was created
- not ready to plan (no pose, no map or no goal)
- replanning

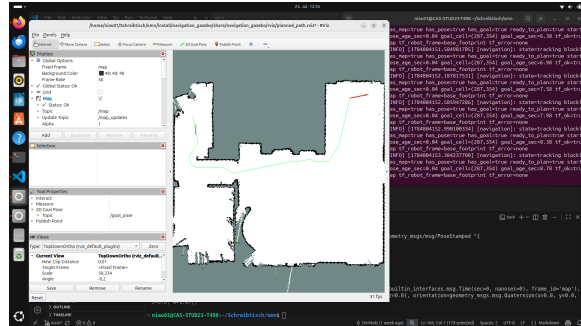
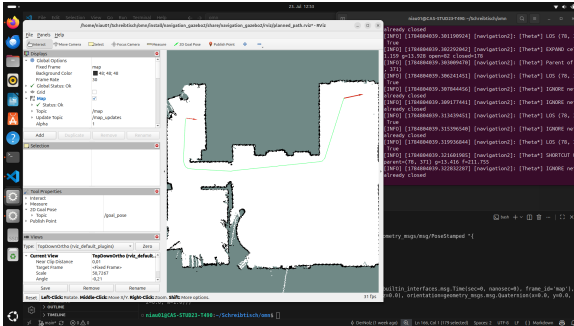
set toggle to true, when:

- following a path

# Theta\* vs A\*

Theta\*

A\*



Time of Computation: approx. 240 Seconds  
Noticably smoother path

Time of Computation: approx. 2 seconds  
Rougher path even after path smoothing

==> A\* wins due to faster computation

## Goal Queue: Problem and Solution

### Previous Behaviour

- Navigation handled only one goal
- New goal replaced the active goal
- Older goals were discarded

### New Behaviour

- Added FIFO goal queue
- Current goal stays active
- New goals wait in queue
- Goals follow received order

### FIFO Order

Goal A  $\rightarrow$  Goal B  $\rightarrow$  Goal C

# Goal Queue: Main Logic

## When a New Goal Arrives

- ➊ Transform the goal into the map frame
- ➋ Check whether it is a duplicate
- ➌ Activate it if no goal is active
- ➍ Otherwise, append it to the queue
- ➎ Reject it if the queue is full

## Queue Settings

- Minimum spacing: 0.25 m
- Maximum size: 100 goals
- New goals added with:

`append()`

- Next goal taken with:

`popleft()`

## Code Added In

- Import section and `__init__()`
- `goal_pose_callback()`
- New `activate_next_goal()` method
- Goal-reached part of `timer_callback()`

# Goal Queue: Simulation Result

## Waypoint Queued

```

navigation-7 [INFO] [178388859.833297231] [navigation]: Planned path with 132 cells; reason:robot off path
navigation-7 [INFO] [178388860.86236082] [navigation]: Got new position
navigation-7 [INFO] [178388866.367987729] [navigation]: Got new position
navigation-7 [INFO] [178388861.699728824] [navigation]: Got new position
navigation-7 [INFO] [178388861.934623883] [navigation]: Got new position
navigation-7 [INFO] [178388862.728160352] [navigation]: Planned path with 131 cells; reason:robot off path
navigation-7 [INFO] [178388862.739682481] [navigation]: Got new position
navigation-7 [INFO] [178388862.211593232] [navigation]: Got new position
rviz2-9 [INFO] [178388863.38456667] [rviz2]: Setting goal pose: Frame:map, Position(13.7475, -7.74699, 0), Orientation(0, 0, 0.596266, 0.8863369) = Angle: 2.9687
navigation-7 [INFO] [178388862.296028192] [navigation]: Queued goal x:13.747, y:-7.746; queue size=3
navigation-7 [INFO] [178388866.350111189] [navigation]: NAVIGATION STATUS | state=MOVING | goal=(5.17, -6.54) | queue=3 | Waypoint added to the navigation queue
navigation-7 [INFO] [178388863.877134339] [navigation]: NAVIGATION STATUS | state=MOVING | goal=(5.17, -6.54) | queue=3 | Following path to active waypoint
navigation-7 [INFO] [178388864.877862953] [navigation]: Got new position
navigation-7 [INFO] [178388864.576573798] [navigation]: Got new position
navigation-7 [INFO] [178388865.419148142] [navigation]: Got new position
navigation-7 [INFO] [178388866.565373239] [navigation]: Got new position
navigation-7 [INFO] [178388866.792473817] [navigation]: Planned path with 129 cells; reason:robot off path
navigation-7 [INFO] [178388866.796182445] [navigation]: Got new position
navigation-7 [INFO] [178388867.229925412] [navigation]: Got new position
navigation-7 [INFO] [178388867.832794193] [navigation]: Got new position
navigation-7 [INFO] [178388868.391624397] [navigation]: Got new position
navigation-7 [INFO] [178388869.267973276] [navigation]: Planned path with 128 cells; reason:robot off path
navigation-7 [INFO] [178388869.277062418] [navigation]: Got new position
navigation-7 [INFO] [178388869.720728178] [navigation]: Got new position
navigation-7 [INFO] [178388870.491743339] [navigation]: Got new position
navigation-7 [INFO] [178388871.212388123] [navigation]: Got new position
navigation-7 [INFO] [178388872.566481323] [navigation]: Planned path with 127 cells; reason:robot off path
    
```

- Queue size reached 3
- Active goal was not replaced

## Queue Completed

```

navigation-7 [INFO] [178388877.584489386] [navigation]: Planned path with 4 cells; reason:robot off path
navigation-7 [INFO] [178388877.511588416] [navigation]: Got new position
navigation-7 [INFO] [178388877.466161230] [navigation]: Got new position
navigation-7 [INFO] [178388877.643636670] [navigation]: Planned path with 3 cells; reason:robot off path
navigation-7 [INFO] [178388877.644618162] [navigation]: Got new position
navigation-7 [INFO] [178388877.26178181] [navigation]: Got new position
navigation-7 [INFO] [178388877.737215585] [navigation]: Waypoint reached x:13.747, y:-7.746
navigation-7 [INFO] [178388877.737799274] [navigation]: NAVIGATION STATUS | state=WAYPOINT_REACHED | goal=(13.75, -7.74) | queue=0 | Reached waypoint (13.747, -7.741)
navigation-7 [INFO] [178388877.736478965] [navigation]: NAVIGATION STATUS | state=END | goal=none | queue=0 | All queued waypoints completed
navigation-7 [INFO] [178388877.738862275] [navigation]: All queued goals succeeded
navigation-7 [INFO] [178388877.739373242] [navigation]: NAVIGATION STATUS | state=GOAL_REACHED | goal=none | queue=0 | Complete waypoint queue executed
navigation-7 [INFO] [178388877.739692372] [navigation]: Got new position
navigation-7 [INFO] [178388877.286449304] [navigation]: Got new position
    
```

- Queue returned to 0
- All queued goals succeeded

## Result

Goals are executed sequentially and are no longer discarded.

## Localization (in Sim)

## Localization (in Sim)

- essentially dynamic obstacle avoidance